INCH-POUND

MIL-M-38510/101G
AMENDMENT 2
02 DECEMBER 1994

SUPERSEDING
AMENDMENT 1
3 MARCH 1989

MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, OPERATIONAL AMPLIFIERS,
MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/101G, dated 8 May 1989, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1/, line 2, delete "dice" and substitute "die".

PAGE 3

3.1, delete and substitute the following;
"3.1 Item requirements. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. The individual item requirements for devices marked with the "Q" certification mark shall be in accordance with MIL-I-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein."

PAGE 4

Table I, add new footnote "1/" after the title which will state the following;
"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

Table I, input offset voltage temperature sensitivity, symbol column, delete "$\frac{\Delta V_{IO}}{\Delta T}$" and substitute "$\Delta V_{IO}/\Delta T$". Also, footnote "1/" will be renumbered as footnote "2/".

Table I, input offset current temperature sensitivity, symbol column, delete "$\frac{\Delta I_{IO}}{\Delta T}$" and substitute "$\Delta I_{IO}/\Delta T$". Also, footnote "1/" will be renumbered as footnote "2/".

Table I, input bias current test, footnote "1/" will be renumbered as footnote "2/". Device types 01, 02, 03, 05, 07, and 08, delete "+1 nA min" and substitute "-0.1 nA min".

PAGE 5

Table I, add new footnote "1/" after the title which will state the following;
"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

Table I, Adjustment for input offset voltage (positive) test, footnote "2/" will be renumbered as footnote "3/".

Table I, Adjustment for input offset voltage (negative) test, footnote "2/" will be renumbered as footnote "3/".

Table I, Output short-circuit current (for positive output) test, footnote "3/" will be renumbered as footnote "4/".

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Table I, output short-circuit current (for negative output) test, footnote "3/" will be renumbered as footnote "4/". Also, delete in its entirety and substitute the following:

Characteristics	Symbol	Conditions (Figure 4, unless otherwise specified)	Limits								Units
			01,02,08		03,05		04,06		07		
			Min	Max	Min	Max	Min	Max	Min	Max	
Output short-circuit current (for negative output)	I_{OS}-	$\pm V_{CC}$ = 15 V, T_A = -55°C to +125°C, t ≤ 25 ms		+60		+60		+15			mA
		T_A = +25°C to +125°C 4/								+65	mA
		T_A = -55°C								+80	mA

Table I, supply current test, footnote "4/" will be renumbered as footnote "5/".

Table I, both open loop voltage gain (single ended) tests, footnote "5/" will be renumbered as footnote "6/". Also, footnote "6/" will be renumbered as footnote "7/".

Table I, Transient response rise time tests, footnote "7/" will be renumbered as footnote "8/".

Table I, Transient response overshoot tests, footnote "7/" will be renumbered as footnote "8/".

Table I, both slew rate tests, footnote "8/" will be renumbered as footnote "9/". Also, footnote "9/" will be renumbered as footnote "10/".

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Table I, settling time test, footnote "10/" will be renumbered as footnote "11/". Also, limits column for device type 07, delete in its entirety and substitute the following:

Characteristics	Symbol	Conditions (Figure 4, unless otherwise specified)	Limits								Units
			01,02,08		03,05		04,06		07		
			Min	Max	Min	Max	Min	Max	Min	Max	
Settling time 11/	t_S+	See figure 5, T_A = +25°C								800	ns
		T_A = -55°C to +125°C								1200	ns
	t_S-	See figure 5, T_A = +25°C								800	ns
		T_A = -55°C to +125°C								1200	ns

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PAGE 6

Add a new footnote "1/" which will state;
"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

Footnote "1/" will be renumbered as footnote "2/".

Footnote "2/" will be renumbered as footnote "3/".

Footnote "3/" will be renumbered as footnote "4/".

Footnote "4/" will be renumbered as footnote "5/".

Footnote "5/" will be renumbered as footnote "6/".

Footnote "6/" will be renumbered as footnote "7/".

Footnote "7/" will be renumbered as footnote "8/".

Footnote "8/" will be renumbered as footnote "9/".

Footnote "9/" will be renumbered as footnote "10/".

Footnote "10/" will be renumbered as footnote "11/".

PAGE 7

4.1, delete and substitute the following;
"4.1 Sampling and inspection. Sampling inspection procedures shall be in accordance with MIL-M-38510 and methods 5004, 5005, and 5007, as applicable, of MIL-STD-883, except as modified herein. For devices marked with the "Q" certification mark, sampling and inspection procedures shall be in accordance with MIL-I-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein."

PAGE 8

4.4.2b, delete the last sentence and substitute the following; "If alternate burn-in conditions are used, the circuit shown in either Figure 7 or Figure 8 shall be used."

4.2.d, line 3, delete "either".

4.4.3.c, lines 3 and 4, delete and substitute and following:
"The LTPD for subgroup 12 shall be 5, and 13 shall be 7 for class B devices."

PAGE 18

FIGURE 4 at the center upper half of the circuit at I_{OS} meter, delete "S_7 and substitute "S_6".
Also, at the center bottom of the circuit S_{4A} position 7, delete the "200 Ω" resistor and substitute "200 kΩ".

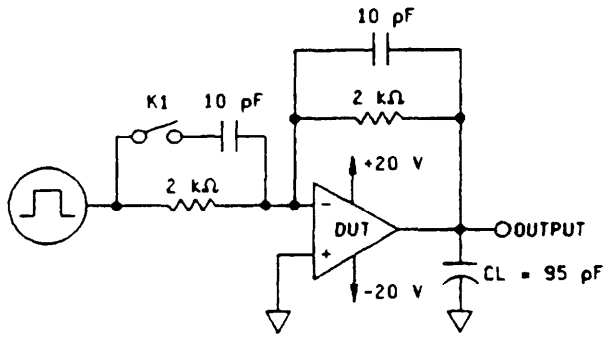
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6/, Delete and substitute the following; "6/ Device type 01, all case types, and device type 02, case A, B, C, and D only."

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FIGURE 5, Add the following diagram as an alternate test circuit for device type 07 only.



NOTES:
1. K1 is closed for small signal tests (Tr and P.O.) and is open for large signal tests (±slew rate, ±Ts).
2. Input signal is a -50 mV to 0 mV pulse train for small signal tests and -5 V to +5 V pulse train for large signal tests.
3. Tr of the input signal is < 10 ns for the small signal tests.

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Figure 5, add note: "9. C_L capacitance specified includes stray, jig, and probe capacitance."

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FIGURE 8. Note 2, delete and substitute the following; "For device types 01 and 04, V_{IN} = 3.0 Vdc. For device type 03, V_{IN} = +3 V or -3 V."

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Table III, test numbers 9-16, device types 01, 02, 03, 05, 07, and 08, delete "+1 nA min" and substitute "-0.1 nA min".

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Table III, test numbers 35-42, device types 01, 02, 03, 05, 07, and 08, delete "+1 nA min" and substitute "-0.1 nA min".

PAGE 32

Table III, test numbers 61-68, device types 01, 02, 03, 05, 07, and 08, delete "+1 nA min" and substitute "-0.1 nA min".

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Table III, test numbers 81 and 83, conditions column, delete "-15 V" and substitute "+15 V".

Table III, test numbers 82 and 84, conditions column, delete "+15 V" and substitute "-15 V".

Table III, test number 86, conditions column, delete "2 kΩ" and substitute "10 kΩ".

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Table III, test numbers 91, 93, 101, and 103, conditions column, delete "-15 V" and substitute "+15 V".

Table III, test numbers 92, 94, 102, and 104, conditions column, delete "+15 V" and substitute "-15 V".

Table III, test number 96 and 106, conditions column, delete "2 kΩ" and substitute "10 kΩ".

Table III, test number 110, conditions column, delete "V_{IN} = -5 V to +5 V" and substitute "V_{IN} = +5 V to -5 V".

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Table III, add a sentence to footnote "1/" which will state the following;
"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

3/, delete and substitute the following; "V_{IN}(adj) is not performed on device type 02, case I only, or on device types 04, 06 and 08, all case types."

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG
Air Force - 80

Preparing activity:
DLA - ES

Agent:
DLA - ES

(Project 5962-1404)

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